

EXPLORER

NAURIN

mongodb-replication-socialmedia

data

scripts

JS checkData.js

JS deleteData.js

JS failoverDemo.js

JS insertActivityLogs.js

JS insertComments.js

JS insertFollowers.js

JS insertPosts.js

JS insertUsers.js

JS setupReplica.js

JS updateData.js

JS insertUsers.js

JS setupReplica.js

JS insertPosts.js

JS insertComments.js

JS insertFollowers.js

mongodb-replication-socialmedia > scripts > JS insertUsers.js > ...

```
1 use("socialmediaDB");
2 db.users.insertMany([
3
4   {
5     userId:"U001",
6     username:"athiya",
7     email:"athiya@gmail.com",
8     bio:"AI Student",
9     followersCount:120,
10    followingCount:80
11  },
12
13  {
14    userId:"U002",
15    username:"john",
16    email:"john@gmail.com",
17    bio:"Developer",
18    followersCount:200
```

JS insertUsers.js ●

JS setupReplica.js

JS insertPosts.js X

JS insertComment.js

mongodb-replication-socialmedia > scripts > JS insertPosts.js > ...

```
1  use("socialmediaDB");
2  db.posts.insertMany([
3
4  {
5    postId:"P001",
6    userId:"U001",
7    caption:"First Post",
8    hashtags:["AI","MongoDB"],
9    likesCount:120,
10   createdAt:new Date()
11  },
12
13  {
14    postId:"P002",
15    userId:"U002",
16    caption:"Vacation",
17    hashtags:["Travel"],
18    likesCount:50,
19    createdAt:new Date()
20  },
21
22  {
23    postId:"P003",
24    userId:"U001",
25    caption:"Learning MongoDB",
```

PROBLEMS

OUTPUT

DEBUG CONSOLE

TERMINAL

PORTS

state: 1,

JS insertUsers.js ●

JS setupReplica.js

JS insertPosts.js

JS insertComments.js

mongodb-replication-socialmedia > scripts > JS insertFollowers.js > ...

```
1  use("socialmediaDB");
2  db.followers.insertMany([
3
4  {
5    followId:"F001",
6    followerId:"U001",
7    followingId:"U002",
8    followedAt:new Date()
9  },
10
11 {
12   followId:"F002",
13   followerId:"U003",
14   followingId:"U001",
15   followedAt:new Date()
16 },
17
18 {
19   followId:"F003",
20   followerId:"U002",
21   followingId:"U003",
22   followedAt:new Date()
23 }
24
25 ]);
```

PROBLEMS

OUTPUT

DEBUG CONSOLE

TERMINAL

PORTS

state: 1,

JS insertUsers.js ●

JS setupReplica.js

JS insertPosts.js

JS insertComments.js ✕

JS insertFollowers.js

mongodb-replication-socialmedia > scripts > JS insertComments.js > ...

```
1  use("socialmediaDB");
2  db.comments.insertMany([
3
4  {
5    commentId:"C001",
6    postId:"P001",
7    userId:"U002",
8    commentText:"Nice post",
9    commentedAt:new Date()
10 },
11
12 {
13   commentId:"C002",
14   postId:"P001",
15   userId:"U003",
16   commentText:"Good Work",
17   commentedAt:new Date()
18 },
19
20 {
21   commentId:"C003",
22   postId:"P003",
23   userId:"U002",
24   commentText:"Interesting",
25   commentedAt:new Date()
```

JS setupReplica.js

JS insertPosts.js

JS insertComments.js

JS insertFollowers.js

mongodb-replication-socialmedia > scripts > JS insertActivityLogs.js > ...

```
1  use("socialmediaDB");
2  db.activity_logs.insertMany([
3
4  {
5    activityId:"A001",
6    userId:"U001",
7    activityType:"Login",
8    device:"Mobile",
9    activityTime:new Date()
10 },
11
12 {
13   activityId:"A002",
14   userId:"U002",
15   activityType:"Post",
16   device:"Laptop",
17   activityTime:new Date()
18 },
19
20 {
21   activityId:"A003",
22   userId:"U001",
23   activityType:"Like",
24   device:"Mobile",
25   activityTime:new Date()
26 },
27 ])
```

PROBLEMS

OUTPUT

DEBUG CONSOLE

TERMINAL

PORTS

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
PS C:\Users\Athiy\naurin\mongodb-replication-socialmedia> mongosh --port 27017 scripts\setupReplica.js
>>
```

Replica Set Initialized

```
PS C:\Users\Athiy\naurin\mongodb-replication-socialmedia> mongosh --port 27017
```

```
>>
```

Current Mongosh Log ID: 6a1474fbda7966a46cab113

Connecting to: **mongodb://127.0.0.1:27017/?directConnection=true&serverSelectionTimeoutMS=2000&appName=mongosh+2.8.3**

=mongosh+2.8.3

Using MongoDB: 8.3.2

Using Mongosh: 2.8.3

For mongosh info see: <https://www.mongodb.com/docs/mongodb-shell/>

The server generated these startup warnings when booting

2026-05-25T21:41:04.211+05:30: Access control is not enabled for the database. Read and write access to databases and configuration is unrestricted

2026-05-25T21:41:04.212+05:30: This server is bound to localhost. Remote systems will be unable to connect to this server. Start the server with --bind_ip <address> to specify which IP addresses it should serve responses from, or with --bind_ip_all to bind to all interfaces. If this behavior is desired, start the server with --bind_ip 127.0.0.1 to disable this warning


```
_id: 0,
name: 'localhost:27017',
health: 1,
state: 1,
stateStr: 'PRIMARY',
uptime: 251,
optime: { ts: Timestamp({ t: 1779725705, i: 1 }), t: Long('1') },
optimeDate: ISODate('2026-05-25T16:15:05.000Z'),
optimeWritten: { ts: Timestamp({ t: 1779725705, i: 1 }), t: Long('1') },
optimeWrittenDate: ISODate('2026-05-25T16:15:05.000Z'),
lastAppliedWallTime: ISODate('2026-05-25T16:15:05.868Z'),
lastDurableWallTime: ISODate('2026-05-25T16:15:05.868Z'),
lastWrittenWallTime: ISODate('2026-05-25T16:15:05.868Z'),
syncSourceHost: '',
syncSourceId: -1,
infoMessage: '',
electionTime: Timestamp({ t: 1779725535, i: 1 }),
electionDate: ISODate('2026-05-25T16:12:15.000Z'),
configVersion: 1,
configTerm: 1,
self: true,
lastHeartbeatMessage: ''
},
{
  _id: 1,
  name: 'localhost:27018',
  health: 1,
  state: 2,
  stateStr: 'SECONDARY',
  uptime: 190,
  optime: { ts: Timestamp({ t: 1779725705, i: 1 }), t: Long('1') },
  optimeDurable: { ts: Timestamp({ t: 1779725705, i: 1 }), t: Long('1') },
```

ents.js

JS insertFollowers.js

JS insertActivityLogs.js

JS checkData.js X

mongodb-replication-socialmedia > scripts > JS checkData.js

```
1  use("socialmediaDB");
2  print("\n===== USERS =====");
3  db.users.find().pretty();
4
5  print("\n===== POSTS =====");
6  db.posts.find().pretty();
7
8  print("\n===== COMMENTS =====");
9  db.comments.find().pretty();
10
11 print("\n===== FOLLOWERS =====");
12 db.followers.find().pretty();
13
14 print("\n===== ACTIVITY LOGS =====");
15 db.activity_logs.find().pretty();
```


ments.js JS insertFollowers.js JS insertActivityLogs.js JS checkData.js JS updateData.js X JS deleteData.js

mongodb-replication-socialmedia > scripts > JS updateData.js > ...

```
1  use("socialmediaDB");
2  db.users.updateOne(
3  {
4  userId:"U001"
5  },
6  {
7  $set:{
8  bio:"AI Engineer and Developer"
9  }
10 }
11 );
12
13 db.posts.updateOne(
14 {
15 postId:"P001"
16 },
17 {
18 $set:{
19 caption:"Learning MongoDB Replication"
20 }
21 }
22 );
23
24 db.posts.updateOne(
25 {
```

ments.js JS insertFollowers.js JS insertActivityLogs.js JS checkData.js JS updateData.js JS deleteData.js X BLAC

mongodb-replication-socialmedia > scripts > JS deleteData.js > ...

```
1 use("socialmediaDB");
2 db.users.deleteOne({
3   followersCount:0
4 });
5
6 db.comments.deleteOne({
7   commentId:"C003"
8 });
9
10 db.posts.deleteOne({
11   postId:"P002"
12 });
13
14 print("\nREMAINING USERS");
15 db.users.find().pretty();
16
17 print("\nREMAINING COMMENTS");
18 db.comments.find().pretty();
19
20 print("\nREMAINING POSTS");
21 db.posts.find().pretty();
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
PS C:\Users\Athiy\aurin\mongodb-replication-socialmedia> mongosh --port 27017 scripts\insertData.js
>>
Error: ENOENT: no such file or directory, open 'C:\Users\Athiy\aurin\mongodb-replication-socialmedia\scripts\insertData.js'
PS C:\Users\Athiy\aurin\mongodb-replication-socialmedia> mongosh --port 27017 scripts\insertUsers.js
>>
users inserted successfully
PS C:\Users\Athiy\aurin\mongodb-replication-socialmedia> mongosh --port 27017 scripts\insertPosts.js
>>
posts inserted successfully
PS C:\Users\Athiy\aurin\mongodb-replication-socialmedia> mongosh --port 27017 scripts\insertFollowers.js
>>
followers inserted successfully
PS C:\Users\Athiy\aurin\mongodb-replication-socialmedia> mongosh --port 27017 scripts\insertComments.js
>>
comments inserted successfully
PS C:\Users\Athiy\aurin\mongodb-replication-socialmedia> mongosh --port 27017 scripts\insertActivityLogs.js
>>
activity_logs inserted successfully
PS C:\Users\Athiy\aurin\mongodb-replication-socialmedia> mongosh --port 27017 scripts\checkData.js
>>

===== USERS =====

===== POSTS =====

===== COMMENTS =====

===== FOLLOWERS =====

===== ACTIVITY LOGS =====
```

```
rs0 [direct: primary] socialmediaDB> show collections
activity_logs
comments
followers
posts
users
```

```
rs0 [direct: primary] socialmediaDB> db.users.find()
```

```
[
  {
    _id: ObjectId('6a1476d258178bb128abc114'),
    userId: 'U001',
    username: 'athiya',
    email: 'athiya@gmail.com',
    bio: 'AI Engineer and Developer',
    followersCount: 120,
    followingCount: 80
  },
  {
    _id: ObjectId('6a1476d258178bb128abc115'),
    userId: 'U002',
    username: 'john',
    email: 'john@gmail.com',
    bio: 'Developer',
    followersCount: 200,
    followingCount: 150
  },
  {
    _id: ObjectId('6a1476d258178bb128abc116'),
    userId: 'U003',
    username: 'sara',
    email: 'sara@gmail.com',
    bio: 'Photographer',
    followersCount: 300,
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
},
{
  _id: 1,
  name: 'localhost:27018',
  health: 1,
  state: 1,
  stateStr: 'PRIMARY',
  uptime: 2157,
  optime: { ts: Timestamp({ t: 1779727648, i: 1 }), t: Long('2') },
  optimeDate: ISODate('2026-05-25T16:47:28.000Z'),
  optimeWritten: { ts: Timestamp({ t: 1779727648, i: 1 }), t: Long('2') },
  optimeWrittenDate: ISODate('2026-05-25T16:47:28.000Z'),
  lastAppliedWallTime: ISODate('2026-05-25T16:47:28.529Z'),
  lastDurableWallTime: ISODate('2026-05-25T16:47:28.529Z'),
  lastWrittenWallTime: ISODate('2026-05-25T16:47:28.529Z'),
  syncSourceHost: '',
  syncSourceId: -1,
  infoMessage: '',
  electionTime: Timestamp({ t: 1779727348, i: 1 }),
  electionDate: ISODate('2026-05-25T16:42:28.000Z'),
  configVersion: 1,
  configTerm: 2,
  self: true,
  lastHeartbeatMessage: ''
},
{
  _id: 2,
  name: 'localhost:27019',
  health: 1,
  state: 2,
  stateStr: 'SECONDARY',
  uptime: 2124,
  optime: { ts: Timestamp({ t: 1779727638, i: 1 }), t: Long('2') },
```